

The reaction quotient, Q (HL)

Higher level (HL)

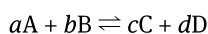
Learning outcomes

By the end of this section you should be able to calculate the reaction quotient, Q , from the concentrations of reactants and products at a particular time and determine the direction in which the reaction will proceed in order to reach equilibrium.

By this stage in the subtopic, you should be getting the idea that reactions can go in both the forward and reverse directions and reach an equilibrium position when the rate of the forward reaction is equal to the rate of the reverse reaction. But what about a reaction that is not yet at equilibrium? Is there a way to determine in what direction a reaction not at equilibrium will proceed to reach its equilibrium position? By measuring the concentration of reactants and products at any time in a reaction, the reaction quotient, Q , can be calculated. Its value is then compared to the value of the equilibrium constant, K_c , and the direction in which the reaction will proceed can then be determined.

The reaction quotient is a measure of the relative amounts of reactants and products for a reaction that is not at equilibrium. It is used to determine in which direction a reaction will proceed to reach equilibrium and to compare to the equilibrium constant to see whether or not the system has reached equilibrium. The equation for the reaction quotient looks similar to that for the equilibrium constant in that it has the concentrations of the products in the numerator and the concentrations of the reactants in the denominator. Both concentrations of the reactants and products are raised to the powers of their coefficients in the balanced equation.

So for the general reaction between reactants A and B to form products C and D, where a , b , c and d are coefficients:

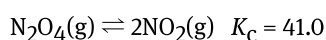


the equation for the reaction quotient is:

$$Q = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

The reaction quotient is calculated using the given concentrations of a system and is used to determine whether a reaction where the concentrations are changing is occurring, and in what direction that reaction is proceeding.

Take for example the reaction shown in which dinitrogen tetroxide, N_2O_4 , reacts to produce nitrogen dioxide, NO_2 .



The value of the equilibrium constant for this reaction indicates that the equilibrium position lies to the right and the equilibrium mixture contains a higher concentration of products than reactants. **Table 1** shows the concentrations of N_2O_4

and NO_2 in a new system, at a particular point in time. We can use these concentrations to calculate the value of the reaction quotient and determine in which direction the reaction will proceed to reach equilibrium.

Table 1. The concentrations of N_2O_4 and NO_2 at a particular time.

	$\text{N}_2\text{O}_4(\text{g})$	$\text{NO}_2(\text{g})$
Concentration at time t (mol dm^{-3})	2.00	6.00

Start by writing the expression for the reaction quotient for the reaction:

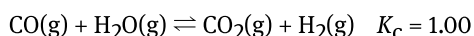
$$Q = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$$

Next, substitute in the concentrations of the reactant and products and calculate the value of the reaction quotient:

$$\begin{aligned} Q &= \frac{[6.00]^2}{[2.00]} \\ &= 18.0 \end{aligned}$$

Now that we have calculated the value of the reaction quotient, what does it tell us about the reaction? In this example the value of the reaction quotient is less than the value of the equilibrium constant. This means that the concentration of products is lower than at equilibrium, which means that the reaction will proceed to the right to reach equilibrium. This has the effect of increasing the concentrations of the products until the reaction reaches equilibrium. At this point the value of the reaction quotient is equal to the value of the equilibrium constant and the concentrations of reactants and products remain constant.

In the next example, we will look at the reaction between carbon monoxide and steam to produce carbon dioxide and hydrogen.



The concentrations of reactants and products at a particular time are shown in **Table 2**.

Table 2. The concentrations of reactants and products in the reaction between carbon monoxide and steam at a particular time.

	$\text{CO}(\text{g})$	$\text{H}_2\text{O}(\text{g})$	$\text{CO}_2(\text{g})$	$\text{H}_2(\text{g})$
Concentration at time t (mol dm^{-3})	0.500	0.500	1.00	1.00

First, write the equation for the reaction quotient:

$$Q = \frac{[\text{CO}_2][\text{H}_2]}{[\text{CO}][\text{H}_2\text{O}]}$$

Next, substitute in the concentrations of the reactant and products and calculate the value of the reaction quotient:

$$\begin{aligned} Q &= \frac{[1.00][1.00]}{[0.500][0.500]} \\ &= 4.00 \end{aligned}$$

In this example, the value of the reaction quotient is greater than the value of the equilibrium constant. This means that the reaction mixture has a higher concentration of products than the equilibrium mixture. Therefore, to reach equilibrium, the reaction will proceed to the left to increase the concentration of the reactants until the reaction reaches equilibrium.

The relationship between the reaction quotient and the equilibrium constant is summarised in **Figure 1**.

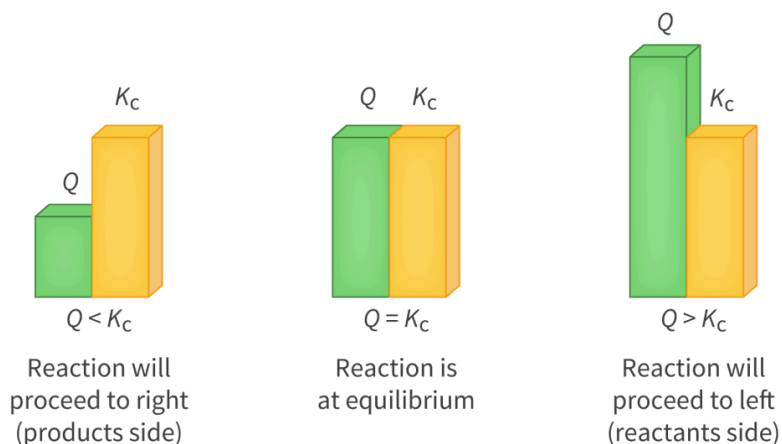


Figure 1. By comparing the value of the reaction quotient, Q , to the value of the equilibrium constant, K_c , for a reaction, the direction in which the reaction will proceed to reach equilibrium can be determined.

 More information for figure 1

The image shows a comparison of the reaction quotient (Q) and the equilibrium constant (K_c) through three bar graphs.

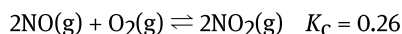
1. The leftmost graph features two bars, with the Q bar being shorter than the K_c bar. Below, it reads: ' $Q < K_c$. Reaction will proceed to right (products side).'
2. The center graph shows two bars of equal height for Q and K_c . The text below states: ' $Q = K_c$. Reaction is at equilibrium.'
3. The rightmost graph has a taller Q bar compared to the K_c bar. Below, it says: ' $Q > K_c$. Reaction will proceed to left (reactants side).'

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- When $Q > K_c$ the concentration of products is greater than at equilibrium. The reverse reaction is favoured and the reaction will proceed to the left to reach equilibrium.
- When $Q = K_c$ the reaction is at equilibrium. The rate of the forward reaction is equal to the rate of the reverse reaction.
- When $Q < K_c$ the concentration of reactants is greater than at equilibrium. The forward reaction is favoured and the reaction will proceed to the right to reach equilibrium.

Worked example 1

Nitrogen monoxide reacts with oxygen to form nitrogen dioxide according to the following equation:



At a particular point in time the concentrations of $\text{NO}(\text{g})$, $\text{O}_2(\text{g})$ and $\text{NO}_2(\text{g})$ are found to be 0.15 mol dm^{-3} , 0.80 mol dm^{-3} and $0.040 \text{ mol dm}^{-3}$ respectively.

Determine the value of the reaction quotient and predict in which direction the reaction will proceed to reach equilibrium.

First, write the equation for the reaction quotient:

$$Q = \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]}$$

Next, input the concentrations of reactants and products and calculate the value of Q.

$$\begin{aligned} Q &= \frac{[0.040]^2}{[0.15]^2[0.80]} \\ &= 0.0888 \dots \\ &= 0.089 \text{ (2 s.f.)} \end{aligned}$$

Finally compare the value of the reaction quotient with the value of the equilibrium constant.

In this example, $Q < K_{\text{c}}$ so the reaction will proceed to the right to reach equilibrium.

Use **Interactive 1** to test your understanding of the value of the reaction quotient, Q, and the direction of reaction.

	The direction of reaction to reach equilibrium	The change in the value of Q to reach equilibrium
$Q > K_{\text{c}}$		
$Q < K_{\text{c}}$		
$Q = K_{\text{c}}$		

The reaction will proceed to the left	The reaction is at equilibrium	The reaction will proceed to the right
Q will increase	Q will not change	Q will decrease

☒ Check

Interactive 1. The reaction quotient, Q, and the direction of reaction.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual activity

Imagine you are attending a ballroom dancing event. The event is a little different because this event is for parents and guardians and their children. A dancing couple is made up of one adult and one child and, as the event goes on, different combinations of adults and children enter the dance floor and then leave the dance floor when they get tired and need to rest. The hall can accommodate a total of 60 adults and 60 children, but the dance floor is only large enough for 50 dancing couples, so the extra dancers must wait for a spot to open up.

1. If we think of this dance event as a system at equilibrium, identify what is required to make this a closed system.
2. Using the equation:



state the equilibrium expression for this system.

3. Deduce the value of the equilibrium constant, K_c , for this system at equilibrium and explain the significance of this value. Imagine that the hall has just opened the doors and the dancers have entered the hall, but the music has not yet begun, so there are no dancers yet on the dance floor. State the value of the reaction quotient, Q , for this scenario and determine to which side the reaction will proceed to reach equilibrium.
4. Describe a scenario where $Q > K_c$ for this analogy and explain how this affects the system.

1. No additional dancers may enter once the hall reaches its capacity of 120 people (60 adults and 60 children).
2. $K = [\text{couple}] / ([\text{child}][\text{adult}])$
3. At equilibrium, there will be 50 couples and the extra dancers waiting. Since a total of 60 adults and 60 children entered, there will be 10 adults and 10 children waiting ($60 - 50 = 10$)

$$K = \frac{50}{10 \times 10} = 0.5$$

The reverse reaction, couples leaving the dance floor, is favoured since $K < 1$.

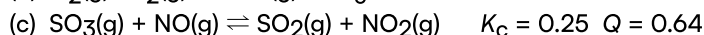
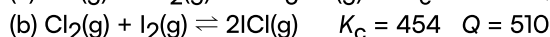
Adults and children will pair up to form a dancing couple until there are exactly 50 couples. The low K value indicates that once equilibrium is reached, the couples will leave the dance floor first and then be replaced by a new couple. Although the dance floor holds exactly 50 dancing couples at equilibrium, it takes more time for an adult and child to form a couple and

enter the dance floor than it does for a tired couple to leave the dance floor and separate.

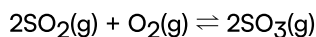
4. $Q = 0$, since there are no couples formed on the dance floor yet (no products). The reaction will proceed to the right/forward to reach equilibrium as adults and children pair up to form a couple.
5. If $Q > K$, then more than 50 couples are on the dance floor (greater concentration of products compared to reactants). This is greater than the number of dancing couples permitted, resulting in more couples leaving the dance floor than those entering. This would cause the reaction to shift to the left/reverse to reach equilibrium.

Now let's try some questions with chemical reactions:

1. For the following reactions, compare the values of Q and K_C and determine to which side the reaction will proceed to reach equilibrium.



2. Calculate the value of the reaction quotient, Q , for the following reaction.



	SO₂(g)
Concentration at time t (mol dm ⁻³)	0.254

The K_C for the reaction was found to be 4.62.

Determine in which direction the reaction will proceed to reach equilibrium.

1. (a) $Q < K_C$ so reaction will proceed to right
 (b) $Q > K_C$ so reaction will proceed to left
 (c) $Q > K_C$ so reaction will proceed to left

2.

$$\begin{aligned}
 Q &= \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]} \\
 &= \frac{[0.500]^2}{[0.254]^2[0.0855]} \\
 &= 45.321 \dots \\
 &= 45.3 \text{ (3 s.f.)}
 \end{aligned}$$

$Q > K_C$ so reaction will proceed to the left to reach equilibrium.